NNN-S-450B
September 27, 1976
SUPERSEDING
NNN-S-00450A
October 3, 1967 and
NNN-S-450
June 23, 1961

FEDERAL SPECIFICATION

SLIDE, MICROSCOPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass microscope slides suitable for laboratory purposes.

1.2 Classification.

1.2.1 Types, sizes, thicknesses and styles. Microscope slides shall be furnished in the following types, sizes, thicknesses and styles, as specified (see 6.1):

Type I. - Plain.

Size 1. - 75 millimeters (mm.) by 25 mm.
Size 2. - 75 mm. by 38 mm.
Size 3. - 3 inches by 1 inch (76.2 mm. by 25.4 mm.).

Thickness A. - 0.75 mm. to 1.10 mm.
Thickness B. - 1.10 mm. to 1.35 mm.

Type II. - With concavities.

Size 1. - 75 mm. by 25 mm.
Size 2. - 75 mm. by 38 mm.

Thickness C. - 1.95 mm. to 2.35 mm.

Style a. - One concavity.
Style b. - Two concavities.

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Type III - Dark field.

Size 1 - 75 mm. by 25 mm.
Size 2 - 75 mm. by 38 mm.

Thickness D - 1.15 mm. to 1.25 mm.
Thickness E - 1.45 mm. to 1.55 mm.
Thickness F - 0.90 mm. to 1.10 mm.

NOTE: Thickness D is for use with Spencer Microscopes; thickness E is for use with Bausch and Lomb Microscopes; and thickness F is for use with Leitz Microscopes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Box, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specifications:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications, Standards and Drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM):

ASTM C162 - Glass and Glass Products, Definition of Terms Relating to.
ASTM E211 - Cover Glasses and Glass Slides for Use in Microscopy, Specifications and Tests for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material shall be clear transparent glass, free from imbedded foreign material, bubbles, seeds, blisters, striae and internal cloudiness.

3.1.1 Color. The faces of the microscope slide shall be colorless when judged by the unaided eye. The slide may possess a slight greenish tint when viewed edgewise, but this coloration shall not materially affect the color of light transmitted through the slide faces.

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3.2 Dimensions.

3.2.1 Length and width. The length and width of the slide shall be as specified in 1.2.1, with a tolerance in each direction of plus or minus 0.8 mm. for the type I slide and plus or minus 0.4 mm. for the type II and type III slides.

3.2.2 Thickness. The thickness shall be within the range specified under thickness (see 1.2.1).

3.2.3 Concavities. Each concavity on the type II slide shall be spherical in shape and highly polished. Concavities shall have a diameter of not less than 15 mm. nor more than 18 mm. The depth of the concavities shall be not less than 0.5 mm. nor more than 0.8 mm.

3.2.3.1 Location of concavities.

3.2.3.1.1 Style a. Style a concavity shall be located in the center of the slide.

3.2.3.1.2 Style b. Style b concavities shall be spaced so that there will be an equal distance of not less than 10 mm. nor more than 15 mm. between the end of the slide and the edge of the concavity at either end.

3.3 Frosted end. When specified, slides shall have one end frosted for a distance of 17 mm. to 21 mm. (see 6.1). Grinding or etching shall be such as to allow pen or pencil writing. If specified, both faces of one end of the slides shall be frosted for a distance of 17 mm. to 21 mm. (see 6.1).

3.4 Precleaning. Unless otherwise specified, type I and type II slides shall be precleaned, and when tested in accordance with 4.4.1.1 and 4.4.1.2, there shall be uniform dispersion of liquid and absence of smears on both surfaces (see 6.1). The precleaned slides shall be free of detergent.

3.5 Boiling. The slides shall show no evidence of fogging, chipping or cracking when tested as specified in 4.4.2.

3.6 Solubility. No pink color shall develop in the test solution when the slides are tested as specified in 4.4.3.

3.7 Workmanship. Workmanship shall be first class throughout. The slides shall be free of defects which detract from appearance or may impair serviceability.

3.7.1 Finish on edges. The slides shall have ground and polished edges that are free of rough and sharp cutting surfaces and corners. The edges may be

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bevelled. The corners shall be square, rounded or bevelled, unless specified that the edges shall be bevelled (see 6.1), or the corners shall be rounded (see 6.1).

3.7.2 Slide surfaces. The slides shall be free of all traces of moisture, cloudiness, detergent and any oily, waxy or syrupy film.

3.8 For military users only.

3.8.1 Date of delivery. The date of delivery shall be not more than 3 months after the date of manufacture, representing the date of unit packaging.

3.8.2 Inspection test date. The inspection test date shall be 24 months from the date of manufacture (see 3.8.1).

3.8.3 Unit of issue.

3.8.3.1 Type I slides. One package containing 72 slides constitutes one unit of issue.

3.8.3.2 Type II slides. One slide, as specified, constitutes one unit of issue.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation with the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with table I and MIL-STD-105. The unit of product for inspection purposes shall be one microscope slide. For visual examination, no more than 10 microscope slides shall be selected from a unit package for type I (see 5.1.1.1), or an intermediate package for type II (see 5.1.1.2). For dimensional examination, no more than 5 microscope slides shall be selected from a unit package for type I or an intermediate package for type II.

TABLE I For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with table II and MIL-STD-105. The unit of product for testing purposes shall be one microscope slide. No more than 5 microscope slides shall be selected from a unit package.

TABLE II For tests

Characteristic	Requirement	Test procedure	Inspection level *
Precleaning--			
slide dip test	3.4	4.4.1.1	S-3
wet blotter test	3.4	4.4.1.2	S-3
Boiling	3.5	4.4.2	S-3
Solubility	3.6	4.4.3	S-3

* Acceptance number shall be zero.

4.3 Examination. The microscope slides shall be examined to determine compliance with all requirements contained in the specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the possible defects listed in the table. Definitions for glass defects listed in the table shall be in accordance with ASTM Designation C162.

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TABLE III Classification of defects

Category	Defect
Major	
101	Glass not clear and transparent.
102	Glass not free from imbedded dirt, foreign material, bubbles, seeds, blisters and striae.
103	Slide not free of cloudiness.
104	Slide not free of oily, waxy and syrupy film.
105	Edges on slide not properly ground and polished.
106	Edges and corners of slide not free of sharp cutting surfaces, or not finished as specified.
107	Slide not free of traces of moisture and adhesion.
108	Slide not free of traces of detergent (i.e. incomplete washing and rinsing)
109	Slide not free of scratches which will affect its use.
110	Slide not free of cracks and breaks.
111	Working edges of slide (i.e., short dimension) not free of chips, uneven edges, ragged edges and rough edges.
112	Concavity on type II slide not properly spherical in shape.
113	Concavity on type II slide not highly polished.
114	End of slide not frosted when and as required.
115	Slide not free of frosted spots beyond designated limits (i.e. present on clear areas).
116	Frosting at end of slide, when required, not completely or evenly applied.
117	Frosting at end of slide, when required, not suitable for application of pen or pencil writing.
Minor	
201	Slide not free of chips at the edges of the long dimensions.
202	Slide not free of uneven, rough and ragged edges on the long dimensions.

4.3.2 Dimensional examination. The slide shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 expressed in terms of percent defective.

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TABLE IV Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

4.3.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-585, PPP-B-601, PPP-B-636, or PPP-B-621, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Precleaning.

4.4.1.1 Slide dip test. The sample slides shall not be touched, handled or fingerprinted on or near the edges and surfaces to be used. The sample slides shall be dipped not more than halfway into a beaker of distilled water, and then examined for proper dispersion of liquid and evidence of smears.

4.4.1.2 Wet blotter test. The sample slides shall not be touched, handled or fingerprinted on or near the edges and surfaces to be used. A wet blotter or wet filter paper shall be rubbed over the slide surfaces. The slides shall be examined for proper dispersion of liquid and evidence of smears.

4.4.2 Boiling test. The slides shall be tested for resistance to boiling in accordance with the applicable requirements of ASTM Designation E211.

4.4.3 Solubility test. The solubility of glass slides shall be tested in accordance with the applicable requirements of ASTM Designation E211.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.1).

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5.1.1 Levels A and B.

5.1.1.1 Unit package.

5.1.1.1.1 Type I. Seventy two slides shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. When full-telescope type of carton is utilized, the bottom of the box shall be scored or otherwise incised to provide a tab which, when lifted into position, shall be capable of inclining and supporting the carton for facilitating withdrawal of slides. A sheet of polymer-coated cellophane shall be placed around the slides in such a manner as to protect the slides and edges, and an additional sheet of polymer-coated cellophane wrap shall be placed around the slides in such a manner as to protect the top and bottom edges prior to being placed in the box. The cellophane wrap shall not have a deleterious effect on the slides. The box shall be lined, coated or treated to prevent deposit of paper fibers on the slides. Each box shall be wrapped in suitable transparent material and be effectively sealed to prevent entry of moisutre.

5.1.1.1.2 Type II. Unit package not required.

5.1.1.2 Intermediate package. Three, five or ten unit packages of type I or thirty six slides of type II, as specified (see 6.1), shall be packed in a snug-fitting box constructed in accordance with PPP-B-636, class domestic. Closure of the box shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Five or ten intermediate packages of type I or thirty two intermediate packages of type II, as specified (see 6.1), shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Five or ten intermediate packages of type I or thirty two intermediate packages of type II, as specified (see 6.1), shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Each unit, intermediate and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each unit, intermediate and exterior container shall be marked as specified in MIL-STD-129. In addition, the unit, intermediate and exterior container shall be marked with type II shelf-life marking as specified in MIL-STD-129 (see 3.8).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Type, size, thickness and style of slide (see 1.2.1).
c. Whether slides with one end frosted are required; whether both faces of one end are required to be frosted (see 3.3).
d. Whether type I or type II slides are not required to be furnished precleaned (see 3.4).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. Quantity required (see 5.1.1.2, 5.2.1 and 5.2.2).

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NNN-S-450B, Slide, Microscope

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